

Notes : Adding and Subtracting Rational Expressions

Warm-Up

1. $\frac{-5x^3y^5}{25x^4y^4}$

$$\frac{-y}{5x}$$

2. $\frac{8(x+3)}{2x} \cdot \frac{x}{x(x+3)}$

$$\frac{4}{x}$$

3. $\frac{x^2+x-2}{5-x} \cdot \frac{x^2-25}{x^2+4x+3}$

$$= \frac{(x+2)(x-1)}{(-1)(x-5)} \cdot \frac{(x+5)(x-5)}{(x+3)(x+1)}$$

$$= \frac{-(x+2)(x-1)(x+5)}{(x+3)(x+1)}$$

When only 2 TERMS → ① Diff of Squares or

4. $\frac{x^2+5x+4}{x^2+2x-8} \div \frac{x^2-1}{x^2-x}$ Flip

$$\frac{(x+4)(x+1)}{(x+4)(x-2)} \cdot \frac{(x)(x-1)}{(x+1)(x-1)}$$

$$= \frac{x}{x-2}$$

5. $\frac{6-x}{x^2+2x+1} \div \frac{x^2-2x-24}{x^2-1}$

$$\frac{-1(x-6)}{(x+1)(x+1)} \cdot \frac{(x+1)(x-1)}{(x-6)(x+4)}$$

$$= \frac{-(x-1)}{(x+1)(x+4)}$$

Notes 8-5

To add or subtract you need to first find the common denominator
 "least common denom" LCD
 = the smallest value that the factors of each denominator can divide into
"evenly".

⇒ we need factored form:

→ LCD = 12

Ex1. $\frac{(3)}{(3)} \frac{1}{4} + \frac{2}{3} \frac{(4)}{(4)}$

$$\frac{3}{12} + \frac{8}{12}$$

$$\frac{11}{12}$$

You! $\frac{(3)3}{(3)5} - \frac{2}{3} \frac{(5)}{(5)}$ LCD: 15

$$\frac{9}{15} - \frac{10}{15}$$

$$-\frac{1}{15}$$

FACTOR IN ORDER TO FIND THE LCD. Worst Case

just multiply
denom's
to get
a common
denom.

Ex2. $\frac{5}{12} + \frac{3}{16}$ $\nearrow$ LCM = $(3)(4)(4)$

$$= \frac{(4)}{(4)} \cdot \frac{5}{(3)(4)} + \frac{3}{(4)(4)} \cdot \frac{(3)}{(3)}$$

$$\frac{20}{48} + \frac{9}{48} = \boxed{\frac{29}{48}}$$

You! $\frac{4}{15} + \frac{3}{25}$

$$\frac{(5)}{(5)} \cdot \frac{4}{(3)(5)} + \frac{3}{(5)(5)} \cdot \frac{(3)}{(3)}$$

$$\frac{20}{75} + \frac{9}{75} = \boxed{\frac{29}{75}}$$

Ex3. $\frac{x}{x-1} + 7x$ $\nearrow$ 1st Write every term as a fraction

$$= \frac{x}{(x-1)} + \frac{7x}{(1)(x-1)}$$

$$= \frac{x + 7x^2 - 7x}{(x-1)} = \boxed{\frac{7x^2 - 6x}{x-1}}$$

You! $\frac{6x}{x+2} + 3$

$$= \frac{6x}{x+2} + \frac{3(x+2)}{(x+2)}$$

$$= \frac{6x + 3x + 6}{(x+2)} = \boxed{\frac{9x+6}{x+2}}$$

or

$$\boxed{\frac{3(3x+2)}{(x+2)}}$$

LCD: $(x+4)^2(x-2)(x+4)(x-2)$

Factor to see if cancels

$$\frac{x(7x-6)}{(x-1)}$$

You! $\frac{(x-3)^5}{(x+2)} - \frac{27}{(x+3)(x+2)}$

Ex4. $\frac{(x+4)^2}{(x-2)(x+4)} - \frac{3}{(x+4)(x-2)}$

$$= \frac{2x+8-3x+6}{(x+4)(x-2)}$$

$$= \boxed{\frac{-x+14}{(x+4)(x-2)}}$$

You! $\frac{(x-3)^5}{(x+2)} - \frac{27}{(x+3)(x+2)}$

$$= \frac{5x-15-2x-4}{(x-3)(x+2)}$$

$$= \boxed{\frac{3x-19}{(x-3)(x+2)}}$$

Ex5. $\frac{x+3}{x-2} + \frac{6x-7}{x^2-3x+2}$

FACTOR!

You! $\frac{x+1}{x-1} + \frac{x+2}{x^2+2x-3}$

$$= \frac{(x-1)(x+3)}{(x-1)(x-2)} + \frac{(6x-7)}{(x-2)(x-1)}$$

$$= \frac{x^2+3x-x-3+6x-7}{(x-1)(x-2)}$$

$$= \frac{x^2+8x-10}{(x-1)(x-2)}$$

Multiply out the numerator

$$\frac{(x+3)(x+1)}{(x+3)(x-1)} + \frac{(x+2)}{(x+3)(x-1)}$$

$$\frac{x^2+4x+3+x+2}{(x+3)(x-1)}$$

D ← "stand in" for denom

$$= \frac{x^2+5x+5}{(x+3)(x-1)}$$

Doesn't factor